

SE2030 - Software Engineering

Week 1 Quiz: Introduction to Software Engineering, Ethics & SDLC Overview
60 Questions | Mix of Multiple Choice (MCQ) and Multiple Select (MSQ)

Instructions: Each question is marked as **[MCQ]** (choose ONE correct answer) or **[Select all that apply]** (choose ALL correct answers). The answer key is provided at the end of this document.

1. What is the primary focus of Software Engineering (SE)?

[MCQ - choose one]

- A) Writing code as fast as possible
- B) Applying engineering principles to build reliable, efficient, and maintainable software
- C) Only testing software before release
- D) Designing hardware components

2. Which of the following is NOT something SE focuses on, according to the lecture?

[MCQ - choose one]

- A) Following structured processes
- B) Ensuring quality at every step
- C) Planning across the entire software lifecycle
- D) Ignoring the software lifecycle to save time

3. Which of the following are benefits of good software engineering?

[Select all that apply]

- A) Saves time and money in the long run
- B) Protects users and ensures their satisfaction
- C) Avoids major failures in critical systems
- D) Guarantees the software will have zero bugs forever

4. Which of these is an example of a critical system mentioned as needing careful software engineering?

[MCQ - choose one]

- A) A simple calculator app
- B) Flight control software
- C) A to-do list app
- D) A wallpaper changer app

5. According to the lecture, why does software engineering matter?

[MCQ - choose one]

- A) It makes coding more complicated
- B) It helps avoid major failures in critical systems like banking and hospital systems
- C) It removes the need for testing
- D) It replaces the need for requirements

6. Which best describes the 'Scope' of Programming compared to Software Engineering?

[MCQ - choose one]

- A) Broad, includes the whole software lifecycle
- B) Narrow, task-oriented
- C) Focused only on deployment

D) Focused only on maintenance

7. In the SE vs Programming comparison, what is the 'End Goal' of Software Engineering?

[MCQ - choose one]

- A) Just working code
- B) Working, reliable, scalable, and user-validated software
- C) Passing a code review
- D) Finishing quickly

8. Which of the following activities are listed as part of Software Engineering?

[Select all that apply]

- A) Analysis
- B) Design and architecture
- C) Coding
- D) Testing, deployment, and maintenance

9. According to the comparison table, how is 'Teamwork' typically handled in Programming vs Software Engineering?

[MCQ - choose one]

- A) Programming requires collaboration; SE is individual
- B) Programming is often done individually; SE requires collaboration and coordination in teams
- C) Both are always done individually
- D) Both always require large teams

10. Which 'Process' style is associated with plain Programming according to the comparison table?

[MCQ - choose one]

- A) Follows structured, well-defined processes (e.g., SDLC)
- B) May be ad hoc or informal
- C) Always follows the Agile framework
- D) Always follows the Waterfall model

11. During what period did 'The Software Crisis' mainly occur?

[MCQ - choose one]

- A) 1930s–1950s
- B) 1960s–1990s
- C) 2000s–2010s
- D) 2015–2020

12. Which of the following were common causes of failure during the Software Crisis?

[Select all that apply]

- A) Over budget
- B) Missed deadlines
- C) System crashes and disasters
- D) Excessive testing

13. What caused the failure of NASA's Mariner 1 rocket, according to the lecture?

[MCQ - choose one]

- A) A missing hyphen in code
- B) A software virus
- C) A hardware malfunction only

D) A user error during launch

14. What was the main reason the Therac-25 failure happened?

[MCQ - choose one]

- A) Reused code without adaptation
- B) Poor testing and missing safety checks
- C) A missing hyphen in code
- D) Wrong code update in a trading bot

15. Which system lost \$440 million in 30 minutes due to a wrong code update, according to the lecture?

[MCQ - choose one]

- A) Therac-25
- B) Ariane 5 Rocket
- C) Knight Capital
- D) Mariner 1

16. What is the Software Development Life Cycle (SDLC) best described as?

[MCQ - choose one]

- A) A single phase of testing
- B) A roadmap for building software the right way
- C) A programming language
- D) A type of database

17. According to the lecture, SDLC helps teams to:

[Select all that apply]

- A) Meet user needs and expectations
- B) Save time and cost by planning ahead
- C) Avoid risks and delivery delays
- D) Skip the requirements phase

18. What is a feasibility study primarily used to determine?

[MCQ - choose one]

- A) The programming language to use
- B) Whether the project should be built at all
- C) The final price of the software
- D) The number of developers needed

19. What is the output of a feasibility study called?

[MCQ - choose one]

- A) Software Requirements Specification (SRS)
- B) Feasibility Report
- C) Test Plan
- D) System Design Document

20. A feasibility study helps decide whether a project is:

[Select all that apply]

- A) Technically possible
- B) Financially affordable
- C) Realistic

D) Popular on social media

21. 'Can we build it with the technology we have?' is the key question of which type of feasibility?

[MCQ - choose one]

- A) Economic Feasibility
- B) Technical Feasibility
- C) Legal Feasibility
- D) Schedule Feasibility

22. 'Will the benefits outweigh the expenses?' relates to which type of feasibility?

[MCQ - choose one]

- A) Technical Feasibility
- B) Economic Feasibility
- C) Operational Feasibility
- D) Schedule Feasibility

23. Checking whether a system complies with GDPR falls under which type of feasibility?

[MCQ - choose one]

- A) Legal Feasibility
- B) Operational Feasibility
- C) Technical Feasibility
- D) Economic Feasibility

24. 'Will users actually use it?' is the key question of which type of feasibility?

[MCQ - choose one]

- A) Schedule Feasibility
- B) Operational Feasibility
- C) Legal Feasibility
- D) Technical Feasibility

25. 'Can we deliver it on time?' relates to which type of feasibility?

[MCQ - choose one]

- A) Schedule Feasibility
- B) Economic Feasibility
- C) Technical Feasibility
- D) Legal Feasibility

26. Which of the following are among the five types of feasibility discussed in the lecture?

[Select all that apply]

- A) Technical
- B) Economic
- C) Legal
- D) Emotional
- E) Operational
- F) Schedule

27. Which phase is the FIRST of the six key SDLC phases introduced in the lecture?

[MCQ - choose one]

- A) Testing
- B) Requirements gathering and Analysis

- C) Deployment
- D) Maintenance

28. Which of the following are among the six key SDLC phases mentioned in the lecture?

[Select all that apply]

- A) Requirements gathering and Analysis
- B) System design
- C) Implementation
- D) Marketing
- E) Testing
- F) Deployment
- G) Maintenance

29. What document is typically produced during Requirements Gathering to clearly capture what the client wants?

[MCQ - choose one]

- A) Feasibility Report
- B) Software Requirements Specification (SRS)
- C) Test Plan
- D) User Manual

30. Who is typically involved in requirements gathering, according to the lecture?

[Select all that apply]

- A) Clients
- B) Users
- C) Business analysts
- D) Regulatory auditors only

31. The Requirements Gathering phase happens right after which phase?

[MCQ - choose one]

- A) Testing
- B) Feasibility Study
- C) Deployment
- D) Maintenance

32. What role does the requirements document play between the client and development team?

[MCQ - choose one]

- A) It acts like a contract outlining what will (and won't) be built
- B) It's just a suggestion with no real value
- C) It replaces the need for testing
- D) It is only used after deployment

33. Which of these are common techniques for gathering requirements, according to the lecture?

[Select all that apply]

- A) Interviews
- B) Questionnaires
- C) Observations
- D) Document analysis
- E) Ignoring stakeholders

- F) Prototyping
- G) Brainstorming

34. A Functional Requirement describes:

[MCQ - choose one]

- A) How the system should perform
- B) What the system should do
- C) The legal boundaries of the system
- D) The cost of the system

35. 'The system should load within 3 seconds' is an example of which type of requirement?

[MCQ - choose one]

- A) Functional Requirement
- B) Non-Functional Requirement
- C) Limitation/Constraint
- D) Stakeholder requirement

36. 'The system must run on Android only' is an example of:

[MCQ - choose one]

- A) Functional Requirement
- B) Non-Functional Requirement
- C) Limitation (Constraint)
- D) Feasibility Study

37. In the Student Attendance Management System example, which statement is the Non-Functional Requirement?

[MCQ - choose one]

- A) The system should allow teachers to mark daily attendance for each student
- B) The system should be accessible on both desktop and mobile devices
- C) The system must be deployed within the university's local network only
- D) None of the above

38. What is a stakeholder, as defined in the lecture?

[MCQ - choose one]

- A) Only the person who pays for the software
- B) Any person or group who has an interest in the software system
- C) Only the developers working on the project
- D) Only government regulators

39. Which of the following are categories of stakeholders mentioned in the lecture?

[Select all that apply]

- A) End Users
- B) Internal Users
- C) External Partners
- D) Regulatory Bodies
- E) Project Sponsor / Client
- F) Competitors

40. Which stakeholder category best describes 'Payment gateways, suppliers, delivery services'?

[MCQ - choose one]

- A) End Users
- B) Internal Users
- C) External Partners
- D) Regulatory Bodies

41. 'Education ministry, data protection authorities (GDPR)' are examples of which stakeholder type?

[MCQ - choose one]

- A) Internal Users
- B) Regulatory Bodies
- C) Project Sponsor
- D) End Users

42. Which of the following are best practices in requirement gathering?

[Select all that apply]

- A) Involve all key stakeholders early in the process
- B) Use clear, simple language, avoiding technical jargon
- C) Validate requirements with stakeholders using reviews or walkthroughs
- D) Guess what users want to save time

43. What is a 'walkthrough' in the context of requirement validation?

[MCQ - choose one]

- A) A final product demo to customers
- B) A peer review meeting where the team goes through requirements, design, or code step by step
- C) A marketing presentation
- D) A type of software test

44. Which of the following are common mistakes in requirement gathering, according to the lecture?

[Select all that apply]

- A) Guessing stakeholder needs
- B) Using too much technical jargon
- C) Ignoring non-functional requirements
- D) Involving all key stakeholders early

45. What is the main purpose of the System Design phase?

[MCQ - choose one]

- A) To write the actual code
- B) To translate the SRS into a system blueprint
- C) To deploy the software
- D) To fix bugs after release

46. What is typically included in the System Design phase?

[Select all that apply]

- A) UI Design
- B) Database Schema
- C) System Architecture
- D) User training manuals

47. What is 'unit testing' as described in the Implementation phase?

[MCQ - choose one]

- A) Testing the entire system at once
- B) Making sure each small part of the system works correctly on its own
- C) Testing only the user interface
- D) A type of feasibility study

48. Which types of testing are mentioned in the Testing phase?

[Select all that apply]

- A) Unit Testing
- B) Integration Testing
- C) System Testing
- D) Acceptance Testing
- E) Feasibility Testing

49. What does Acceptance Testing check?

[MCQ - choose one]

- A) If small pieces of code work correctly
- B) If different modules work well together
- C) If the system meets the client's requirements
- D) If the code compiles without errors

50. What typically happens during the Deployment phase?

[MCQ - choose one]

- A) The finished software is installed in the real environment where users will use it
- B) The requirements are gathered for the first time
- C) The code is written module by module
- D) The system undergoes a feasibility study

51. Which of the following happen during the Deployment phase?

[Select all that apply]

- A) A pilot release may be used to test with a small group first
- B) Users are trained, and user manuals are prepared
- C) Help desks or technical support are set up
- D) The software requirements are finalized for the first time

52. When does Software Maintenance begin?

[MCQ - choose one]

- A) Before requirements gathering
- B) After the software is deployed
- C) During the design phase
- D) During testing only

53. Fixing a crash that happens when users try to log in is an example of which type of maintenance?

[MCQ - choose one]

- A) Adaptive Maintenance
- B) Corrective Maintenance
- C) Perfective Maintenance
- D) Preventive Maintenance

54. Updating an app to support a new version of Android or iOS is an example of:

[MCQ - choose one]

- A) Corrective Maintenance
- B) Adaptive Maintenance
- C) Perfective Maintenance
- D) Legal Maintenance

55. Adding an online payment option to an existing school system is an example of:

[MCQ - choose one]

- A) Corrective Maintenance
- B) Adaptive Maintenance
- C) Perfective Maintenance
- D) Operational Maintenance

56. Which of the following are key ethical values in Software Engineering mentioned in the lecture?

[Select all that apply]

- A) Responsibility
- B) Transparency
- C) Privacy
- D) Fairness
- E) Profitability

57. What does IEEE mainly focus on, according to the lecture?

[MCQ - choose one]

- A) Computer science education only
- B) Hardware, electrical, and computer engineering
- C) Legal compliance only
- D) Marketing standards

58. What does ACM stand for and primarily focus on?

[MCQ - choose one]

- A) Association for Computing Machinery; computer science, education, and software research
- B) American Council for Machinery; hardware manufacturing
- C) Association of Certified Managers; project management
- D) Association for Computer Marketing

59. What is the 'Software Engineering Code of Ethics and Professional Practice' also known as?

[MCQ - choose one]

- A) The ACM-only code from 2018
- B) A joint IEEE/ACM code created in 1999 for software engineers
- C) A code created exclusively by the BCS
- D) A legal requirement in all countries

60. Which organizations/codes are mentioned as providing ethical guidelines relevant to software engineering?

[Select all that apply]

- A) IEEE Code of Ethics
- B) ACM Code of Ethics
- C) BCS Code of Conduct (UK)

- D) Australian Computer Society (ACS) Code of Ethics
- E) IFIP Code of Ethics
- F) FIFA Code of Conduct

Answer Key

1. B
2. D
3. A, B, C
4. B
5. B
6. B
7. B
8. A, B, C, D
9. B
10. B
11. B
12. A, B, C
13. A
14. B
15. C
16. B
17. A, B, C
18. B
19. B
20. A, B, C
21. B
22. B
23. A
24. B
25. A
26. A, B, C, E, F
27. B
28. A, B, C, E, F, G
29. B
30. A, B, C
31. B
32. A
33. A, B, C, D, F, G

- 34. B**
- 35. B**
- 36. C**
- 37. B**
- 38. B**
- 39. A, B, C, D, E**
- 40. C**
- 41. B**
- 42. A, B, C**
- 43. B**
- 44. A, B, C**
- 45. B**
- 46. A, B, C**
- 47. B**
- 48. A, B, C, D**
- 49. C**
- 50. A**
- 51. A, B, C**
- 52. B**
- 53. B**
- 54. B**
- 55. C**
- 56. A, B, C, D**
- 57. B**
- 58. A**
- 59. B**
- 60. A, B, C, D, E**